Uncertainty Quantification of Underwater Acoustic Transmission Loss

Stochastic Bellhop: Monte Carlo · Delta/FOSM · Polynomial Chaos Expansion · 3-Parameter Lognormal

4 Scenarios · 3 Input Distributions · 3 Uncertain Parameters

Problem Setup

Sonar performance depends on Transmission Loss (TL), which is uncertain due to variability in **source frequency** (f), **source depth** (z_S), and **sound velocity profile** (SVP).

Goal: Efficiently propagate input uncertainty through the Bellhop ray-tracing model and estimate $\mathbb{E}[\mathrm{TL}]$, $\mathrm{Var}[\mathrm{TL}]$, and the shadow probability $P(\mathrm{TL} < \mathrm{FOM})$.

Scenarios	deep, shallow, upslope, downslope
Distributions	Normal 1%, 5%, 10%
Parameters	f, z_S, SVP
MC budget	$N = 1000$ LHS + $N = 1000$ IID per combo
Total runs	$4 \times 3 \times 3 \times 2 \times 1000 = 72\,000$

Input Uncertainty Model

Each parameter is perturbed independently:

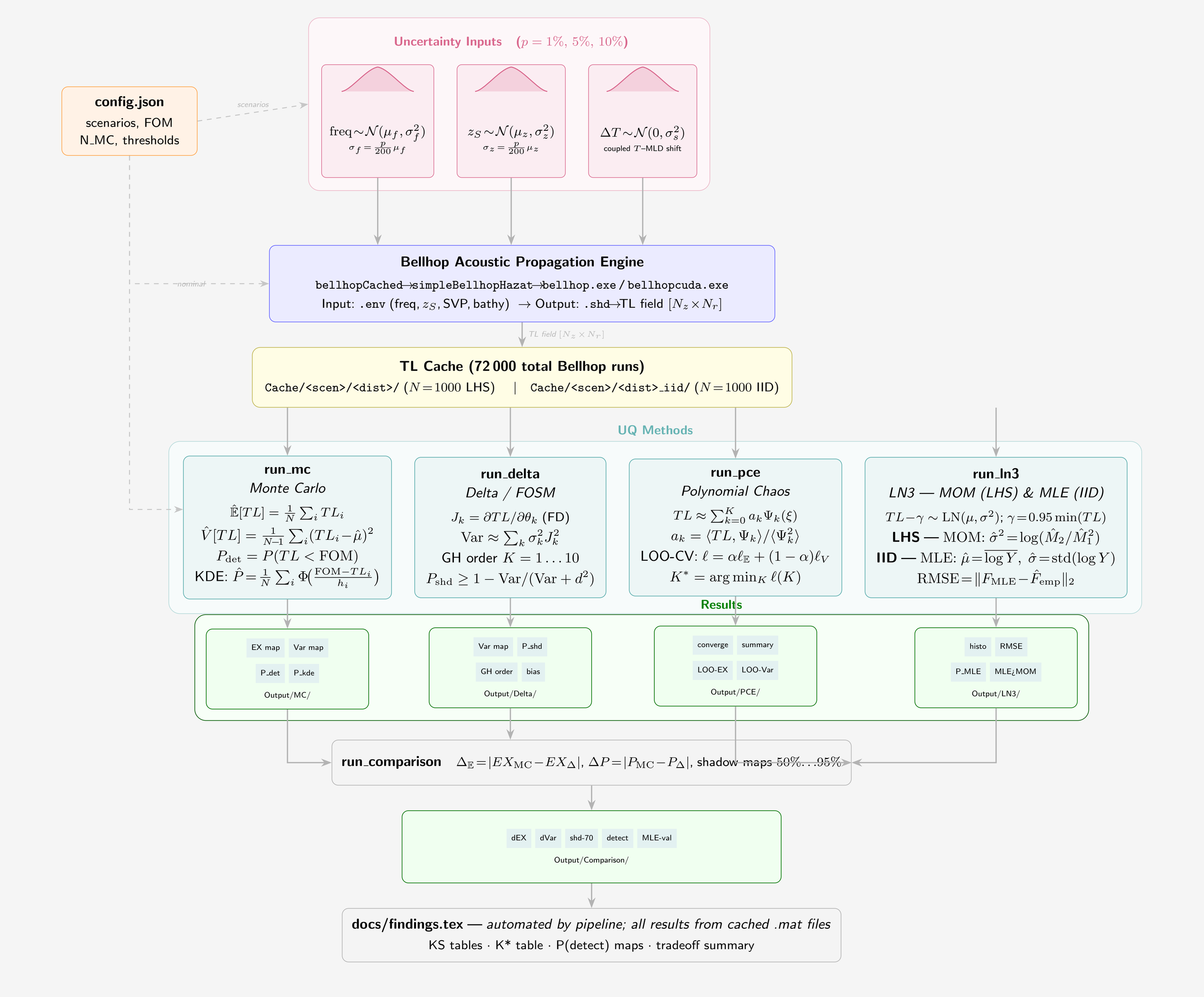
$$f \sim \mathcal{N}\Big(\mu_f,\ \big(\tfrac{p}{200}\mu_f\big)^2\Big)$$

$$z_S \sim \mathcal{N}\Big(\mu_z,\ \big(\tfrac{p}{200}\mu_z\big)^2\Big)$$

$$\Delta T \sim \mathcal{N}(0, \sigma_s^2) \quad \text{(SVP shift)}$$

where $p \in \{1, 5, 10\}$% sets the perturbation magnitude. Samples generated via Latin Hypercube Sampling (LHS).

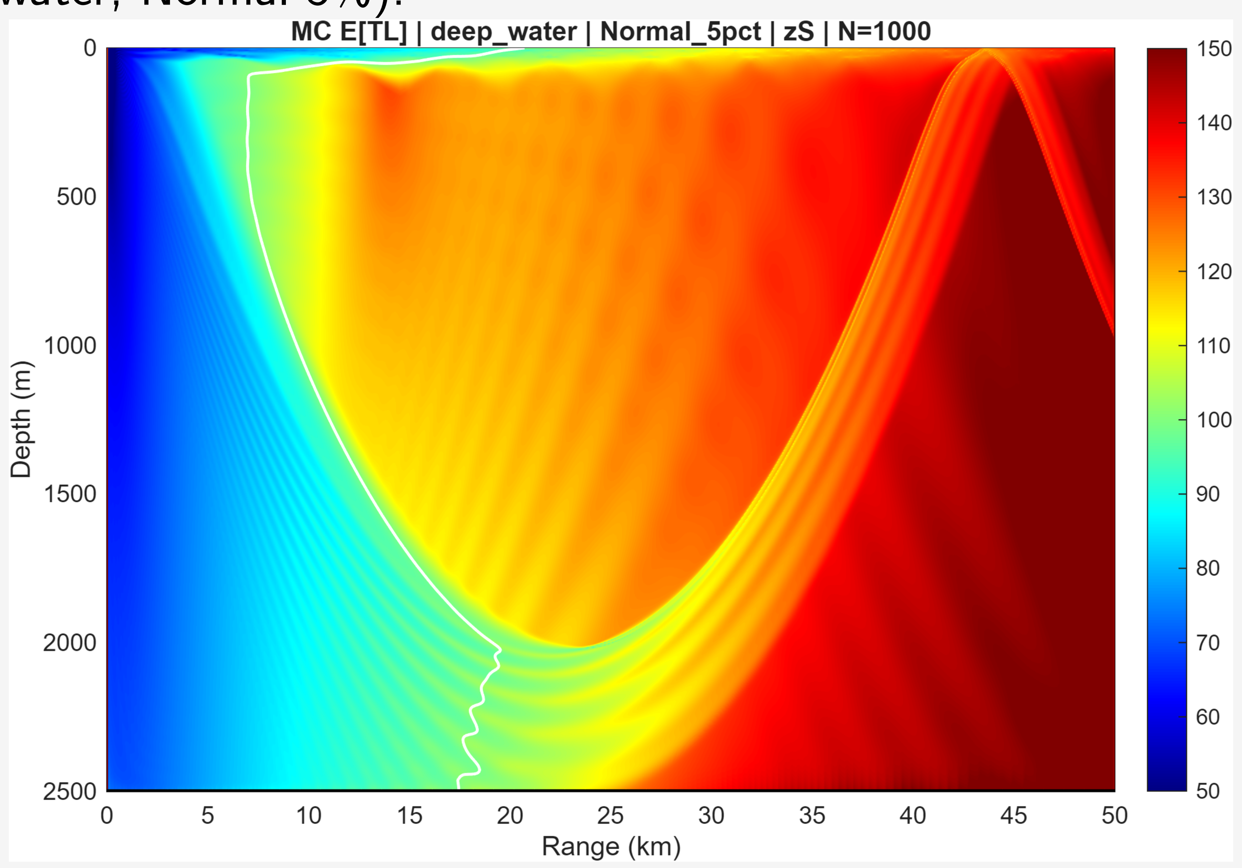
Pipeline Overview



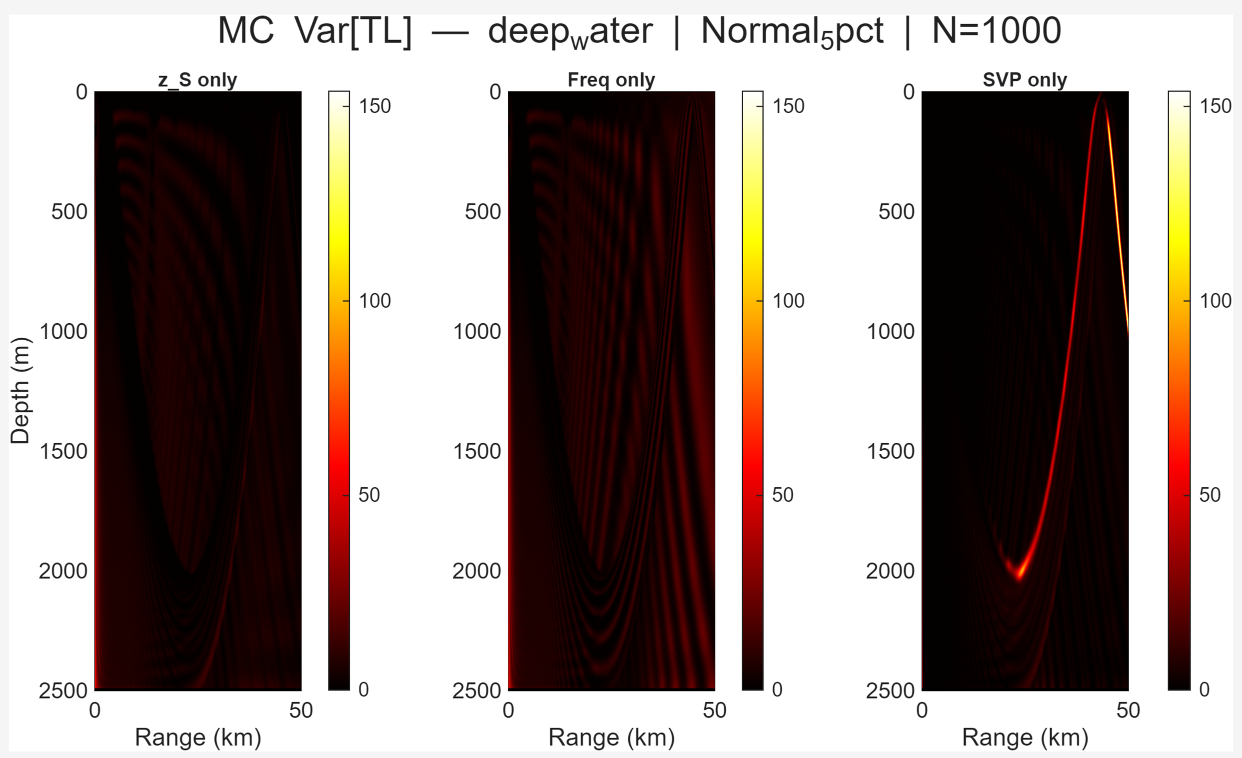
Monte Carlo Reference ($N = 1000$)

The empirical estimators $\hat{\mathbb{E}}[\mathrm{TL}] = \frac{1}{N}\sum_i TL_i$ and $\hat{\mathrm{Var}}[\mathrm{TL}] = \frac{1}{N-1}\sum_i (TL_i - \hat{\mu})^2$ serve as the **gold standard**.

Mean TL and variance under z_S perturbation (deep water, Normal 5%):

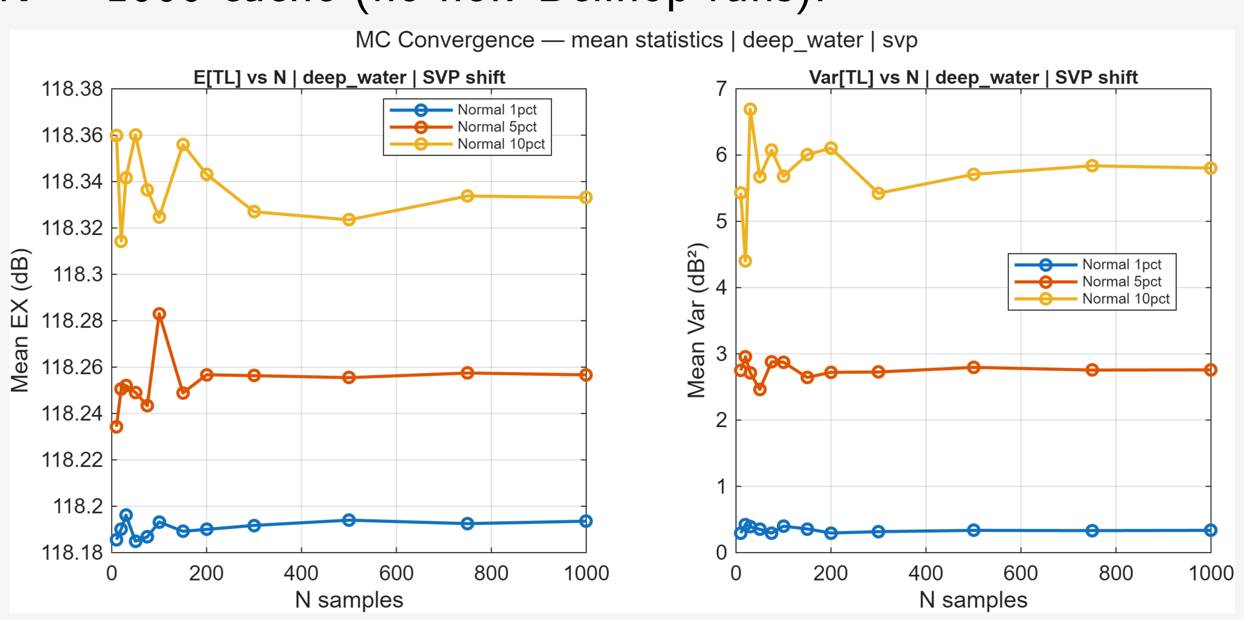


Variance maps for all three parameters:



MC Estimator Convergence

Both $\hat{\mathbb{E}}$ and $\hat{\mathrm{Var}}$ computed from sub-samples of the $N = 1000$ cache (no new Bellhop runs):



Mean stabilises by $N \approx 50$; variance requires $N \approx 200$–500.

Delta / FOSM Method

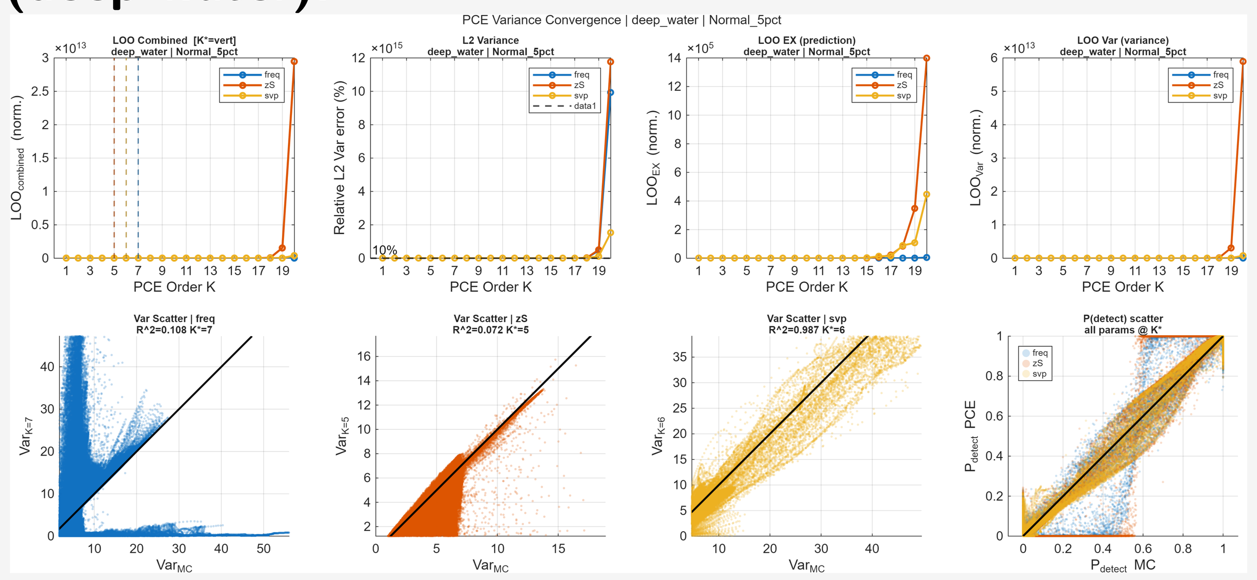
Polynomial Chaos Expansion (PCE)

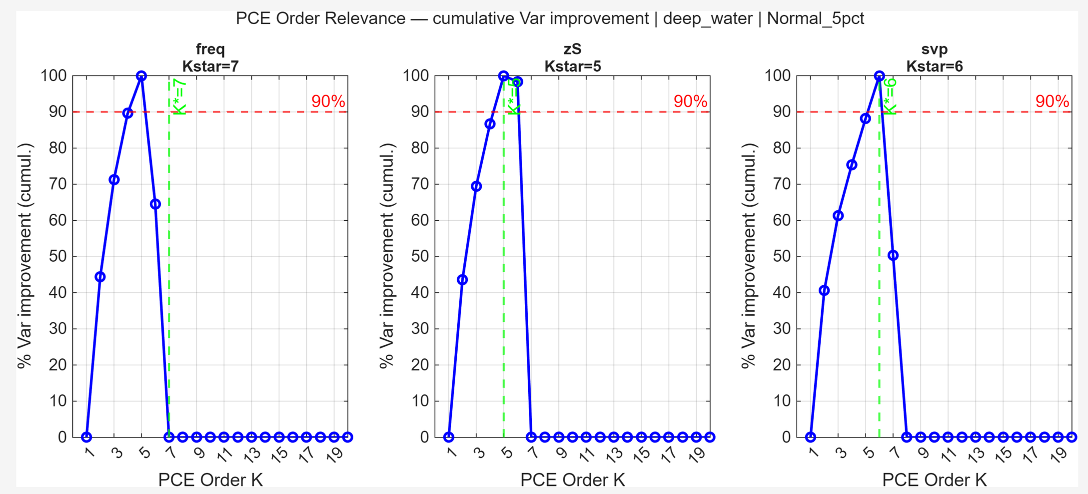
TL is approximated as a polynomial in the standardised input ξ:

$$TL \approx \sum_{k=0}^{K^*} a_k \Psi_k(\xi), \qquad a_k = \frac{\langle TL, \Psi_k \rangle}{\langle \Psi_k^2 \rangle}$$

where Ψ_k are Hermite polynomials. The optimal order K^* is selected by **leave-one-out cross-validation** (LOO-CV) on the combined mean and variance error.

LOO-CV convergence and PCE order relevance (deep water):





Selected orders K^* (all 12 combos):

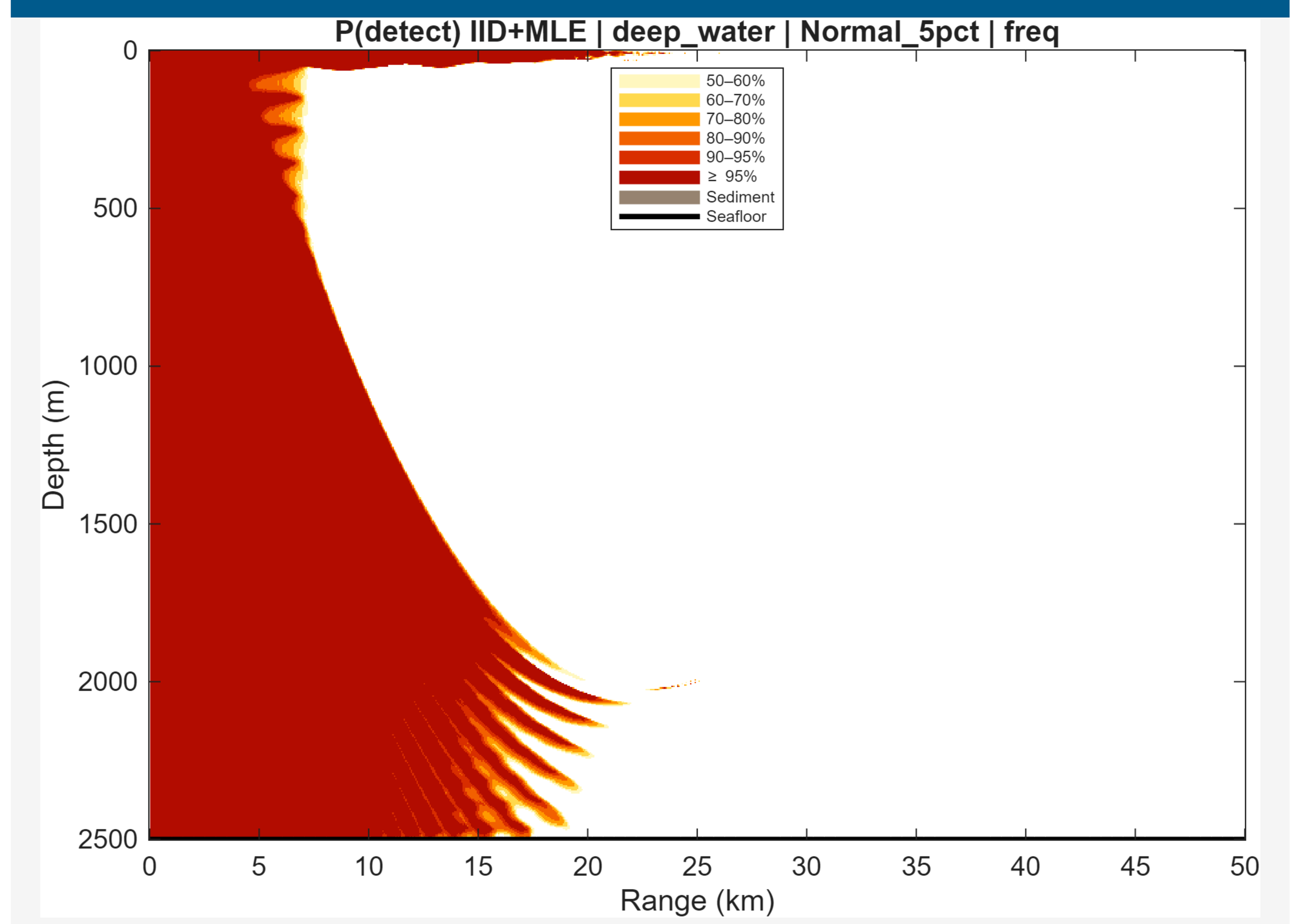
Scenario / Dist	freq	z_S	SVP
deep / 1%	7	6	6
deep / 5%	7	5	6
deep / 10%	7	5	6
shallow / 1%	8	9	7
shallow / 5%	7	5	8
shallow / 10%	7	5	6
upslope / 1%	8	5	5
upslope / 5%	10	5	7
upslope / 10%	12	6	7
downslope / 1%	7	6	5
downslope / 5%	8	6	7
downslope / 10%	11	6	7

3-Parameter Lognormal (LN3) — MLE Fitting

Each pixel's TL distribution is fitted as $TL - \gamma \sim \mathrm{LN}(\mu, \sigma^2)$ with heuristic shift $\gamma = 0.95\min(TL)$. Parameters (μ, σ) are estimated via **Maximum Likelihood** (`lognfit`). Shadow probability: $P(\mathrm{TL} < \mathrm{FOM}) = \Phi_{\mathrm{LN}}(\mathrm{FOM} - \gamma; \mu, \sigma)$.

KS goodness-of-fit (RMSE heatmap, deep water):

Detection Probability: Three Estimators



Estimator	Type	Use
Empirical MC	exact (N=1000)	gold standard
MLE-LN3	parametric	smooth maps
Chebyshev	upper bound	conservative

Method Tradeoff Summary

Method	Cost	When to use
Delta	3 extra FD runs	$\sigma \le 1\%$, flat bathy
PCE (K^* via LOO)	0 extra	σ = 5–10%, flat bathy
LN3-MLE	0 extra	distributional shape, P-maps
MC ($N = 1000$)	1000 runs	slopes, z_S, large σ

Key Findings

1. **Delta is near-exact at 1%** perturbation for flat bathymetry; fails at slope boundaries where the TL response is non-smooth.
2. **PCE converges with K^*=5–9** for flat scenarios; reaches K^*=10–12 for slope/large-σ cases, signalling the edge of polynomial surrogacy.
3. **LN3-MLE is reliable** (KS <0.05) across most of the range–depth grid; bimodal interference zones require MC.
4. **z_S uncertainty is hardest to approximate**: the shadow-zone boundary shifts non-linearly with depth, requiring full MC even at 5% perturbation in deep water.